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CSS for layout: Box model 30m

Let's consider properties related to how the web browser lays out a web page according to the CSS style sheet used in the HTML document. A core concept of layout is the box model.

The box model

The CSS box model is a set of rules that define how the layout is organised by the browser. At the centre of this is the idea that every HTML element is a box. These boxes are packed within the browser window according to the CSS box model. You can use CSS to influence this packing to define a range of layouts.

A CSS box model consists of:

- content – the content that will appear in the box
- padding – the area around the content
- border – the border that appears around the content and padding
- margin – the area around the outside of the box.

It's important to know that each HTML element can have padding and a margin applied to it to change the spacing around or inside the boxes. The overall size of an element is related to the sum of all the layers of the box model, giving the total height and width.

The slideshow below describes the layers of the box model in greater detail.

Scroll through the slideshow using the arrows < > at either side.
Skip to main content

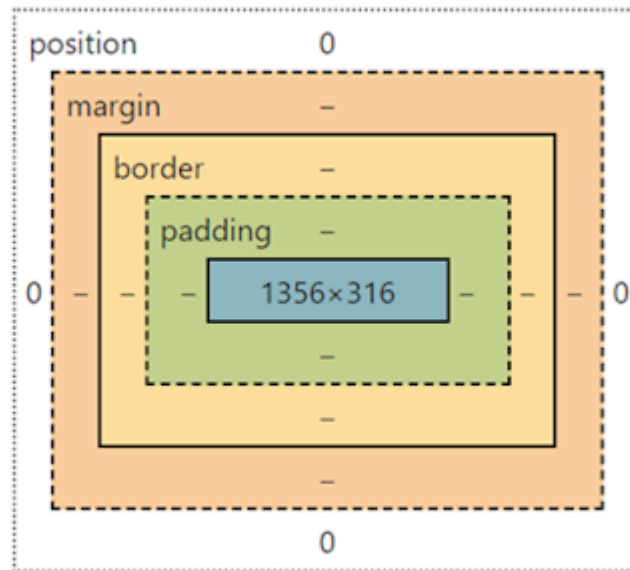
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Image 1 of 5: The central blue box is the content. It's surrounded by padding, border and margin layers. Position determines where the element is on the page.

© Screenshot © DevTools



You can access the F12 web developers panel to find the live CSS box model. The Box Model in DevTools (Codecademy, n.d.)^{fnref:9cd177} explains how to do this.

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How do you use the box model?

Watch *Learn CSS box model in 8 minutes* (Web Dev Simplified, 2019) (8:21)^{fnref:943890} for a practical run-down of using the CSS box model. Be sure to use YouTube's pause and repeat features to pace the video if the presenter is moving a bit too quickly.



ENTER FULLSCREEN

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So, from that video, you learned how to specify the padding, border and margin within your style sheets.
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For example, if you wanted to create a box with a blue border that had extra area for padding and margins, then you'd use CSS code like the example below.



HTML

CSS

Result

EDIT ON

LIVE

```
div {
  background-color: lightgrey;
  border: 15px solid blue;
  padding: 50px;
  margin: 20px;
  text-align: center; /* Optional:
Center the content inside the div */
}

h1 {
  color: navy;
  font-family: Arial, sans-serif;
}

p {
  color: darkslategray;
  font-family: Georgia, serif;
}
```

Resources1x0.5x0.25xRerun

Example 6.7 Boxes



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Element height and width

You should also consider the height and width of your box size. When setting the height and width, you will use either absolute or relative length units.

	Absolute length units	Relative length units
Definition	Remain the same size and do not change based on other elements. These length units are commonly used in print material.	Are relative to something else. Since they are relative, they will scale relative to everything else on the page.
Example	centimetres (cm), pixels (px), inches (in)	em (relative to the current font size), viewport (vh and vw)
Note	We need to be careful when we use these units, because these boxes will not change when the web page is displayed on different screen sizes.	100 vh and 100 vw are values equating to 100% viewport height and 100% viewport width.

The viewport is the part of the web page currently visible on the monitor. This is different from the traditional 100% height and 100% width, which are 100% of the container and could be very small or large. Obviously, you can use any percentage number such as 50 vh or 33 vw. These would equate to 50% of the viewport height and 33% of the viewport width, respectively.



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HTML

CSS

Result

EDIT ON

LIVE

```
.box {  
  width: 1000px;  
  height: 100px;  
  padding: 10px;  
  border: 5px solid gray;  
  margin: 0px;  
  background-color: lightgrey;  
}
```

Resources1x0.5x0.25xRerun

Example 6.8 Viewport



☒ Relative length units

Non-Assessed

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Using the code presented in the CodePen example above, change the height and width so the box uses relative length units that scale relative to other elements and the size of the screen. You want your box to scale as the size of the screen changes in size.

☐ I have successfully completed the task!

Submit



Dig deeper

To learn more about the CSS box model, explore these resources:

- CSS Box Model (W3Schools, n.d.)^{fnref:54a7c6} gives a basic description of what the box model is and how you can work out height and width using a calculation.
- The Box Model (MDN, 2025)^{fnref:3044b0} for more in-depth information about the box model, including block and inline boxes, examples of using the box model and more advanced styling features of each property.
- CSS Values and Units (MDN, 2025)^{fnref:48338e} if you want to explore the different length value units available in CSS.



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Now that we've explored basic styling, let's dive into CSS page layout techniques to structure and position content effectively on a web page.



Thoughts on this page?

